

Bayesian Machine Learning in Quantitative Finance

Chapter 14: Conclusions

Based on Mongwe, Mbuva & Marwala

Outline

- 1 The Whole Book at a Glance
- 2 14.1 Summary of Contributions
- 3 14.2 Ongoing and Future Research Directions
- 4 Key Takeaways

Big-Picture Recap: Chapters 3–13, One Table I

Intuition

Chapter 1 opened with a coin flip and four open questions (explainability, parameter uncertainty, principled model selection, input relevance). Chapter 2 gave us the computational engines (MCMC and variational inference). Every chapter since has been the same Bayesian recipe – prior $\times$ likelihood, normalized by the evidence – applied to a new corner of quantitative finance. Before we summarize contributions and future work, here is the whole book on one page.

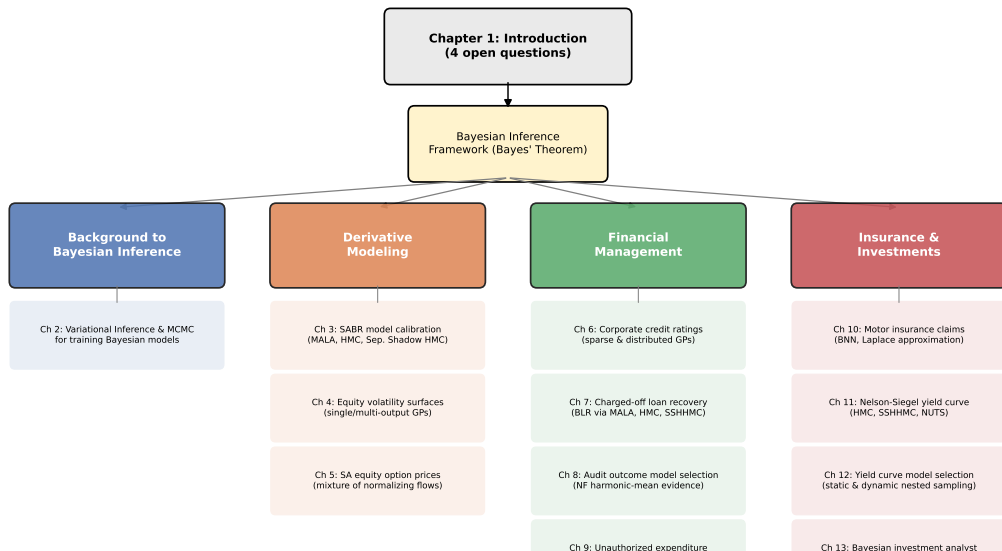
Ch.	Application	Bayesian ML method(s)	Theme
2	Background: training & evaluating Bayesian models	Variational inference, MCMC	Background
3	SABR model calibration (interest-rate skew)	MALA, HMC, Separable Shadow HMC	Derivative modeling
4	Equity volatility surfaces (SA, Europe)	Single-/multi-output Gaussian processes	Derivative modeling
5	European option prices (JSE, emerging mkt)	Mixture of normalizing flows	Derivative modeling
6	Corporate credit ratings	Sparse & distributed (product-of-experts) GPs	Financial management
7	Recovery on charged-off loans	Bayesian logistic regr. via MALA, HMC, SSHMC	Financial management
8	Audit opinion model selection	NF harmonic-mean evidence (MALA/HMC/NUTS)	Financial management
9	Unauthorized municipal expenditure	BLR-ARD via MH, MALA, HMC, Magnetic HMC	Financial management
10	Motor insurance claims likelihood	Bayesian neural net + Laplace approx., ARD	Insurance & investments
11	Nelson–Siegel yield curve calibration	HMC, Separable Shadow HMC, NUTS	Insurance & investments
12	Yield curve model selection	Static & dynamic nested sampling (evidence)	Insurance & investments
13	Bayesian investment analyst (JSE)	BLR-ARD via MCMC (multiple samplers)	Insurance & investments

Big-Picture Recap: Chapters 3–13, One Table II

Notice the pattern: every row is the same story – a point-estimate machine learning problem, re-cast so that the posterior (over parameters, over model choice, or over an entire function) is the object of interest. That is the thread this chapter ties together.

Big-Picture Recap: Chapters 3–13, One Table III

The Whole Book, One Page: 14 Chapters, Four Themes



The Bayesian ML Toolkit, Chapter by Chapter

Intuition

Across 11 application chapters, the book only ever reached for a handful of computational tools. Recognizing *which* tool fits *which* problem shape is one of the most transferable skills from this book.

The Bayesian ML Toolkit: Which Chapter Used What

Metropolis-Hastings						✓				
MALA	✓				✓	✓				✓
Hamiltonian Monte Carlo (HMC)	✓				✓	✓	✓		✓	✓
Separable Shadow HMC	✓				✓				✓	✓
Magnetic HMC						✓				
No-U-Turn Sampler (NUTS)						✓			✓	✓
Sparse / Distributed GPs		✓		✓						
Normalizing Flows			✓			✓				
Bayesian NN + Laplace approx.							✓			

14.1 Summary of Contributions: The Book's Own Framing

What this book set out to do (in the authors' own words)

The advent of artificial intelligence and data-driven methods is transforming quantitative finance. This book explored various domains in quantitative finance and presented the theory and case studies of the potential of machine learning in this space. Employing the Bayesian inference framework – and consequently Bayesian machine learning – lets us naturally answer:

- ① How can we **explain** the prediction or output of a model?
- ② What is the **distribution** of the model's parameters?
- ③ How do we **select between** different models in a statistically principled manner?
- ④ Which **inputs** are most relevant for the task at hand?

Remember Chapter 1? These are exactly the four open questions that motivated the whole book. The framework was applied across *derivative modeling*, *banking*, *financial management*, *insurance*, and *investments* domains.

Theme 0 – Background to Bayesian Inference

What the book says

For this theme, we explored the Bayesian paradigm and how this translates to Bayesian machine learning. We discussed the advantages and disadvantages of the Bayesian framework, how Bayesian models can be trained and deployed, and the performance metrics used to assess them. We further explored how the framework applies across derivative modeling, banking, financial management, insurance, and investments, and highlighted the impact machine learning is having on the finance industry.

Remember Chapter 2? This is precisely where MALA, HMC, NUTS, Separable Shadow HMC, and Magnetic HMC were introduced as the workhorse samplers that every later application chapter reused.

Theme 1 – Derivative Modeling: Contributions I

SABR model calibration (Chapter 3)

We calibrated the Stochastic Alpha Beta Rho model with MALA, HMC, and Separable Shadow Hamiltonian Hybrid Monte Carlo. The approach precisely prices the interest-rate volatility skew on both simulated and ZAR swaption data, with the added benefit of a *distribution* over model predictions and parameters. The MCMC methods are reasonably confident around the at-the-money point but less confident deep in/out of the money, and more confident on the skew's right wing than its left wing – useful for generating volatility stress scenarios.

Equity volatility surfaces (Chapter 4)

Single- and multi-output Gaussian processes, applied to equity volatility surface data from emerging (South Africa) and developed (Europe) markets, calibrate the rough Bergomi and Heston stochastic volatility models. Accuracy is similar to deep neural networks, with the added benefit of uncertainty across the surface and fewer hyperparameter configurations. Calibration becomes more specific (narrower confidence bands) at shorter maturities and at lower relative moneyness.

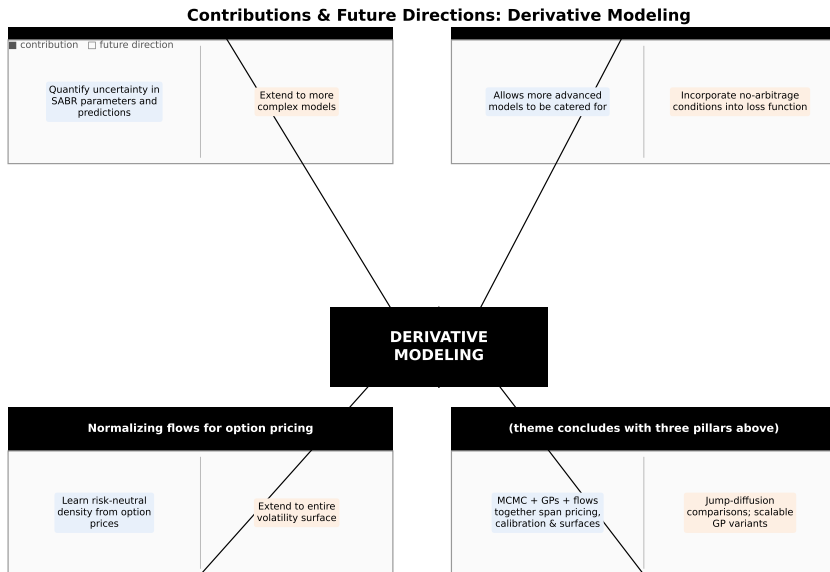
Theme 1 – Derivative Modeling: Contributions II

Option prices via normalizing flows (Chapter 5)

We priced European equity options in an emerging-market context with an entirely data-driven approach based on normalizing flows – learning the risk-neutral density directly (which already imposes no-arbitrage conditions) rather than bolting external arbitrage constraints onto the loss function. A mixture of normalizing flows reproduces option prices across maturities and moneyness levels; options close to expiry require more mixture components, and the largest pricing errors occur at low strikes.

Remember Chapters 3–5? Three different tools (MCMC, GPs, normalizing flows), one shared payoff: every derivative-modeling result in this book comes with a distribution, not just a point price.

Theme 1 – Derivative Modeling: Contributions III



Theme 2 – Financial Management: Contributions I

Corporate credit ratings (Chapter 6)

Sparse and distributed Gaussian processes – variations of the product-of-experts distributed GP – predict corporate credit ratings and scale favorably to large datasets compared with classical GPs. The *debt ratio*, *profit margin-related* metrics, and *return on capital ratio* are the most important financial metrics driving corporate credit rating.

Recovery on charged-off loans (Chapter 7)

A Bayesian logistic regression model, trained with MALA, HMC, and Separable Shadow HMC, detects whether recoveries occur on non-performing loans. The *purpose* of the loan, *home-ownership* status, and the *interest rate* charged are key features. The framework can be extended to more complex models such as Bayesian deep neural networks.

Theme 2 – Financial Management: Contributions II

Audit outcome model selection (Chapter 8)

For municipal audit opinions, we compared Bayesian logistic regression models with varying numbers of covariates using the **Bayesian evidence**, estimated via a harmonic-mean estimator whose target distribution is learned with normalizing flows (fed by MALA/HMC/NUTS samples). The evidence computed on training data reliably predicts which model will outperform on unseen data, with a positive correlation to the Area Under the ROC Curve.

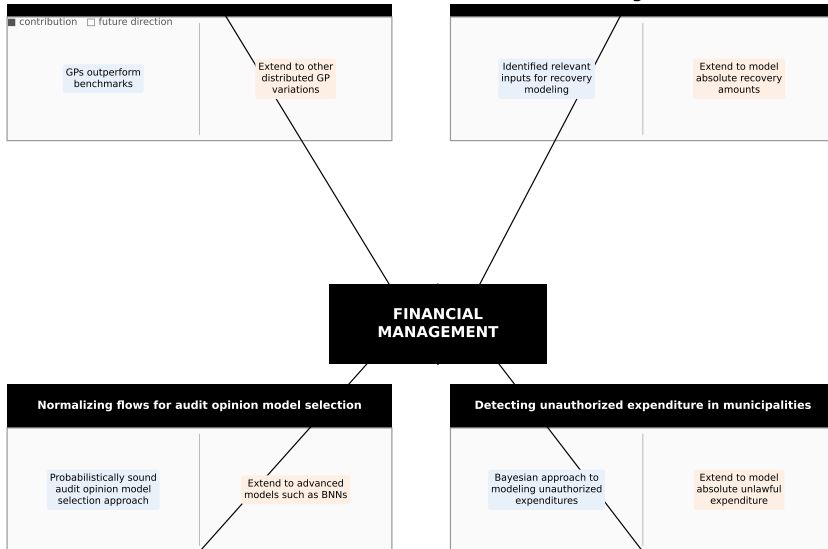
Unauthorized expenditure (Chapter 9)

A Bayesian logistic regression model with an **automatic relevance determination (BLR-ARD)** prior, trained with Metropolis-Hastings, MALA, HMC, and Magnetic HMC, automatically ranks the input features most relevant to detecting unauthorized expenditure in South African municipalities – a tool auditors can use for risk-rating municipal accounts, extendable to irregular and wasteful expenditure.

Remember Chapters 6–9? The evidence metric from Chapter 2's toolbox reappears here as the statistically principled way to *select* between models – the third of the book's four open questions, answered concretely.

Theme 2 – Financial Management: Contributions III

Contributions & Future Directions: Financial Management



Theme 3 – Insurance and Investments: Contributions I

Motor insurance claims (Chapter 10)

We modeled the likelihood of motor insurance claims using Bayesian neural networks trained with the Laplace (Gaussian) approximation of MacKay (1992) – a variational inference variant. Deeper architectures outperform wider ones, and the Bayesian evidence selects the architecture that generalizes best. An ARD prior identifies *claims history*, *type of fuel*, *power of the vehicle*, *lapse*, and *number of policies* as the essential features – making an otherwise black-box neural network interpretable to non-expert stakeholders.

Nelson–Siegel yield curve calibration (Chapter 11)

We calibrated the Nelson & Siegel (1987) yield curve model with HMC, Separable Shadow Hybrid HMC, and NUTS. The calibration gives comparable or better yield curve estimates than ordinary least squares across multiple performance metrics, with a full account of parameter uncertainty.

Theme 3 – Insurance and Investments: Contributions II

Yield curve model selection (Chapter 12)

We used the **nested sampling** evidence estimation method of Skilling (2004) to select between nested Nelson–Siegel models, and introduced the **ARD-Nelson-Siegel (ARD-NS)** model, trained with NUTS, which automatically ranks the short-, medium-, and long-term factor loadings. The evidence framework reliably detects the model that generated the data, and the ARD-NS model shows that yield curve dynamics are driven mainly by the *medium-term (curvature)* factor loading.

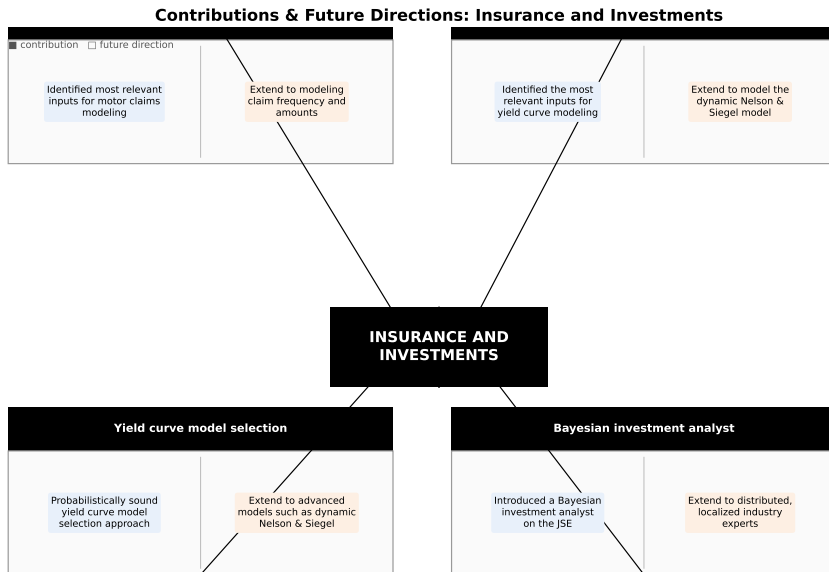
Bayesian investment analyst (Chapter 13)

A BLR-ARD model trained with MCMC methods builds a Bayesian sell-side investment analyst for JSE-listed stocks. It has the best performance on industrial stocks, then resources, then financial stocks, and outperforms the mean consensus of sell-side analysts across all four experiments considered. *Relative volatility* and *initial price* matter most for the full JSE universe; the important features vary meaningfully by sector.

Theme 3 – Insurance and Investments: Contributions III

Remember Chapters 10–13? BNNs (Chapter 10) and nested sampling (Chapter 12) are the two tools we had not yet used elsewhere in the book – both introduced precisely because insurance claims and yield curve model selection needed them.

Theme 3 – Insurance and Investments: Contributions IV



14.2 Ongoing and Future Research Directions

Setting the scene (in the authors' own words)

This book has highlighted various problems in quantitative finance where the Bayesian framework produced good insights. However, many domains within quantitative finance – and other models and algorithms – still need to be explored to extend the work presented in this manuscript. A summary of ongoing and future research across the themes covered in this book follows.

Why this matters for you: every future direction below is a natural “next chapter” you could now write yourself, using tools already in your hands from Chapters 2–13.

Future Directions – Derivative Modeling I

SABR calibration (extends Chapter 3)

Enhance by considering the *entire* interest-rate volatility surface, rather than limiting data to the 1W-into-1Y and 6M-into-1Y swaptions used in this book. Compare against other models that capture the volatility skew, such as jump-diffusion models (Mongwe, 2015; Mongwe et al., 2022).

Volatility surface GPs (extends Chapter 4)

Improve by ensuring the inducing points and kernels used for the multi-output GP are *not* shared. All models in this book approximated the pricing function without explicitly incorporating no-arbitrage conditions (Sidogi et al., 2022) – future work will incorporate them explicitly. A key limitation is the computational complexity of GPs; state-of-the-art scaling approaches, such as distributional GPs (Cohen et al., 2020), are a planned remedy.

Normalizing flows for option pricing (extends Chapter 5)

Compare various techniques for *optimally selecting the number of mixture components* in the normalizing flow mixture. It could also be interesting to jointly model the *entire* option price surface (across all expiries and strikes) instead of the volatility skew per expiry (Yang & Hospedales, 2023).

Remember Chapter 4's discussion of no-arbitrage conditions? This is precisely the limitation future work aims to close, on both the GP and normalizing-flow sides of the derivative modeling theme.

Future Directions – Financial Management I

Credit ratings via distributed GPs (extends Chapter 6)

Expand by considering other distributed GPs, such as a *mixture* of experts instead of a *product* of experts. Data from a developing nation such as South Africa could also assess model performance, and selecting the *number of experts* for the distributed GP is itself an open problem (Cohen et al., 2020).

Loan recovery modeling (extends Chapter 7)

Extend to more complex models such as deep neural networks. Rather than only detecting *whether* recoveries are present, modeling the *actual absolute recovery amount* on the defaulted loan using Bayesian techniques is an open problem, still unexplored in detail in the literature.

Future Directions – Financial Management II

Audit opinion model selection (extends Chapter 8)

Extend to more complicated models, such as deep neural networks, and more advanced sampling techniques based on Riemannian manifolds (Marwala et al., 2023; Mongwe et al., 2021b). Compare the model selection results against other Bayesian evidence estimation approaches, such as nested sampling (Skilling, 2004; Speagle, 2020).

Unauthorized expenditure (extends Chapter 9)

Expand to model the *absolute value* of unauthorized expenditure, and jointly model other unlawful spending such as *fruitless and wasteful expenditure*. Explore more complex models, such as Bayesian neural networks, for this task.

Remember Chapter 12's nested sampling? Notice it resurfaces here as a proposed *alternative* evidence-estimation method for the audit-opinion model-selection problem of Chapter 8 – the book's own future work cross-pollinates between its themes.

Future Directions – Insurance and Investments I

Motor insurance claims (extends Chapter 10)

Scale the Bayesian neural network to larger models and select between different models/architectures. Consider other BNN training methods, such as MCMC, that guarantee convergence to the true target posterior. Instead of only forecasting *whether* a claim occurs, extend to model the *absolute value* of claims through analysis of claim frequency and severity.

Nelson–Siegel yield curve (extends Chapters 11–12)

Extend the analysis to the **dynamic Nelson–Siegel model**, which captures yield curve dynamics over time, and to modeling the *term structure of volatilities* using the Nelson–Siegel model (promising initial results in Guo et al., 2014). For model selection, incorporate yield curve models beyond variations/subsets of Nelson–Siegel, and try other evidence-calculation methods, such as normalizing flows and the harmonic mean estimator (Polanska et al., 2023).

Future Directions – Insurance and Investments II

Bayesian investment analyst (extends Chapter 13)

Consider more complicated models, such as Bayesian neural networks, instead of Bayesian logistic regression (Mbuva et al., 2021). Training *experts in each sector* and aggregating results could significantly improve performance – potentially approached with distributed GP models such as the product-of-experts (Cohen et al., 2020).

Other themes not covered in this book

Machine learning has also been used for: hedging options (deep hedging, Buehler et al., 2019), predicting stock prices from limit order book data (Sidogi et al., 2023), and generating synthetic financial markets data (Sidogi et al., 2022). These and other problems are ripe for framing within the Bayesian framework.

Remember Chapter 6's product-of-experts GP? It is proposed again here, in Chapter 13's future work, as a way to build *distributed, localized industry experts* for the investment analyst – the same tool, a new application.

Closing Words of the Book

In the authors' own words

As highlighted above and throughout this book, there are various avenues that one can explore to enhance and extend the work presented across the different themes in this book. We look forward to the increased deployment of Bayesian inference frameworks in practice within the quantitative finance domain.

Remember the coin flip from Chapter 1? Every idea in this book – SABR skews, credit ratings, audit opinions, motor claims, yield curves, JSE stock calls – was that same coin-flip logic, applied at scale: start with a belief, observe data, update the belief, and keep the resulting *uncertainty* rather than throwing it away.

Key Takeaways: One Idea per Method Family

- **Bayesian thinking (Ch. 1–2):** treat unknowns as distributions, not single numbers – the posterior $\mathbb{P}(\theta \mid \mathcal{D}) \propto \mathbb{P}(\mathcal{D} \mid \theta)\mathbb{P}(\theta)$ is the object of interest, always.
- **MCMC / HMC family (MALA, HMC, Separable Shadow HMC, Magnetic HMC, NUTS – Ch. 3, 7–9, 11, 13):** when there is no closed-form posterior, simulate samples from it; gradient information (Hamiltonian dynamics) makes sampling efficient in high dimensions.
- **Gaussian processes (Ch. 4, 6):** put a Bayesian belief over an entire *function* (a volatility surface, a rating function), and use sparsity/distribution tricks to make that tractable at scale.
- **Normalizing flows (Ch. 5, 8):** learn flexible, invertible probability distributions directly from data – useful both for pricing (learning a risk-neutral density) and for estimating hard normalizing constants (the evidence).
- **Bayesian neural networks (Ch. 10):** a distribution over network weights turns a black box into an uncertainty-aware, and via ARD, an *interpretable* predictor.
- **Nested sampling / the evidence (Ch. 8, 12):** the evidence $\mathbb{P}(\mathcal{D}) = \int \mathbb{P}(\mathcal{D} \mid \theta)\mathbb{P}(\theta) d\theta$ is not just a normalizing constant – it is a principled, automatic way to compare and select between competing models.
- **Automatic relevance determination (Ch. 6, 9, 10, 12, 13):** a prior can do double duty as a feature-selection mechanism, ranking which inputs actually drive the prediction.
- **The unifying thread:** every application in this book answers the same four questions from Chapter 1 – explainability, parameter uncertainty, principled model selection, and input relevance.